33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F22-SR-001
415V CARBON TREATMENT & ELUTION SWITCHROOM

F22-TR-031
TRANSFORMER ROOM

F22-TR-031
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

4000/ 1A
2.5VA
5P20

F22-SW-031
CARBON TREATMENT & ELUTION 415V MOTOR CONTROL CENTRE No.2

1/700A
NOTE 1
4000/ 5A
5VA
Cl 1
4000/ 5A
30VA
Cl 0.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
ACB
AFD
4000A
50
51
50G
51G
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6

F22-SW-131
EMERGENCY MCC
2000-F22-SLD-EL-10261

415V 3NE~ 50Hz 4000A 65kA for 1sec

03F-11 630A 250A EOL
04F-13 630A 250A EOL 3PX F22-KN-411I 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 1 2000-F22-SCM-EL-10260
04F-11 630A 250A EOL 3PX F22-KN-411J 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 2 2000-F22-SCM-EL-10261
02F-11 630A 250A EOL 3PX F22-KN-411K 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 3 2000-F22-SCM-EL-10262
04F-01 630A 250A 3PX F22-KN-411L 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 4 2000-F22-SCM-EL-10263
01F-11 630A 250A EOL 3PX F22-KN-411M 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 5 2000-F22-SCM-EL-10264
01R-01 INC 630A 250A 3PX F22-KN-411N 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 2 BANK 6 2000-F22-SCM-EL-10265
04R-17 400A 400A
400A
FEEDER TO F22-SW-131
EMERGENCY MCC
2000-F22-SCM-EL-10271
02R-11 630A 250A EOL 3PX F22-KN-4110 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 3 BANK 1 2000-F22-SCM-EL-10266
01R-13 630A 250A 3PX F22-KN-411P 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 3 BANK 2 2000-F22-SCM-EL-10267
03R-13 630A 250A 3PX F22-KN-411Q 168kW CIL4 CARBON REGENERATION KILN HEATERS ZONE 3 BANK 3 2000-F22-SCM-EL-10274
04R-09 400A 400A CP 137.8kW F22-CU-406 KILN MERCURY ABATEMENT CHILLER 2000-F22-SCM-EL-10232
E-SHUT F22SR001-LCS NOTE 12

CONTINUED ON SHT 2
2000-F22-SLD-EL-10252

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F22-SW-030 INCOMER TIER." CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF THE SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F22-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET
- 3PX — THYRO POWER CONTROLLER THYRO-PX-3PX-500-350-H-F WITH TOUCH DISPLAY OPERATED IN TAKT MODE (ZERO CROSS FIRING)

NORTHERN STAR

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F22-SLD-EL-10251_Rev1

CIL4 carbon treatment & elution F22-SW-031 415V MCC single line diagram sheet 1